

Project Proposal Form

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DockAgent: An LLM-Driven System for Automated Dockerfile Generation, Testing, and Flakiness Repair

Project Description

DockAgent is a VS Code tool that automatically generates Dockerfiles for software repositories and refines them until the environment builds and runs successfully. It also creates tests to validate container behavior and can test both newly generated and existing Dockerfiles. Additionally, it detects and fixes unstable or flaky builds, ensuring the final Docker setup is both correct and reliable.

Problem Statement & Objectives

Setting up Docker environments for software projects is often a difficult and time-consuming process that requires developers to manually configure dependencies, select appropriate base images, and ensure that the system builds and runs correctly. Even when a Dockerfile initially works, it can later break due to dependency changes or environment differences, making reproducibility and stability hard to maintain.

The objective of **DockAgent** is to automate this process by generating correct and reliable Dockerfiles, validating container behavior through automated tests, and detecting as well as fixing build instability. It aims to reduce manual effort and make Docker based environments more reliable and consistent.

Proposed Solution

- **Dockerfile Generation**

The system then iteratively builds and refines the Dockerfile based on build and runtime feedback, before optimizing the final configuration for stable and successful application execution inside containers.

The system then iteratively generates prompts to guide the LLM in refining and patching the initially generated Dockerfile until it executes successfully, while also optimizing the final Docker image size by reducing unnecessary layers and dependencies.

- **Test Generation**

Analyze Docker image layers to identify outputs such as files, binaries, and environment variables, and generate tests to verify container correctness. This module follows the layer-based analysis approach proposed in [\[1\]](#) and uses Google's Container Structure Tests (CST) framework to validate the generated Docker images.

- **Flakiness Detection & Repair**

Perform repeated builds and analyze build logs to detect unstable or inconsistent behavior caused by dependency changes, network issues, or environment inconsistencies. This component is inspired by the characterization and repair techniques proposed in [\[2\]](#), combining static and dynamic analysis with automated fixes such as version pinning and base image adjustments.

- **Unified Feedback Loop**

Combine test execution feedback (failures, missing dependencies) and flakiness feedback (unstable builds) into a single iterative loop. The system continuously updates the Dockerfile, rebuilds, and revalidates until both correctness and stability are achieved.

- **VS Code Integration**

Provide a seamless developer experience with one click Dockerfile generation, automated testing, inline debugging suggestions, and a chat-based assistant for explaining errors and recommending fixes.

Key Technologies & Tools

- **Languages:** TypeScript, Python
- **Containerization:** Docker
- **AI/Agent Framework:** LLM (OpenAI / local)
- **Knowledge Retrieval:** RAG with ChromaDB
- **Backend:** FastAPI
- **Testing:** Container Structure Tests (CST)
- **IDE Integration:** VS Code Extension API
- **Version Control:** Git, GitHub

References:

[1] Yuki Goto, Shinsuke Matsumoto and Shinji Kusumoto, "[Toward Automated Test Generation for Dockerfiles Based on Analysis of Docker Image Layers](#)" EASE '25: Proceedings of the 29th International Conference on Evaluation and Assessment in Software Engineering, 2025

[2] Taha Shabani, Noor Nashid, Parsa Alian and Ali Mesbah, "[Dockerfile Flakiness: Characterization and Repair](#)"

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Signature of the supervisor: _____

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Proposal Presentation Feedback:

Midterm Presentation Feedback: